Lung Injury

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Chest Trauma

Chest trauma

1. Rib fractures

The most common type of chest trauma usually involves the fifth through the ninth ribs.

Chest Trauma

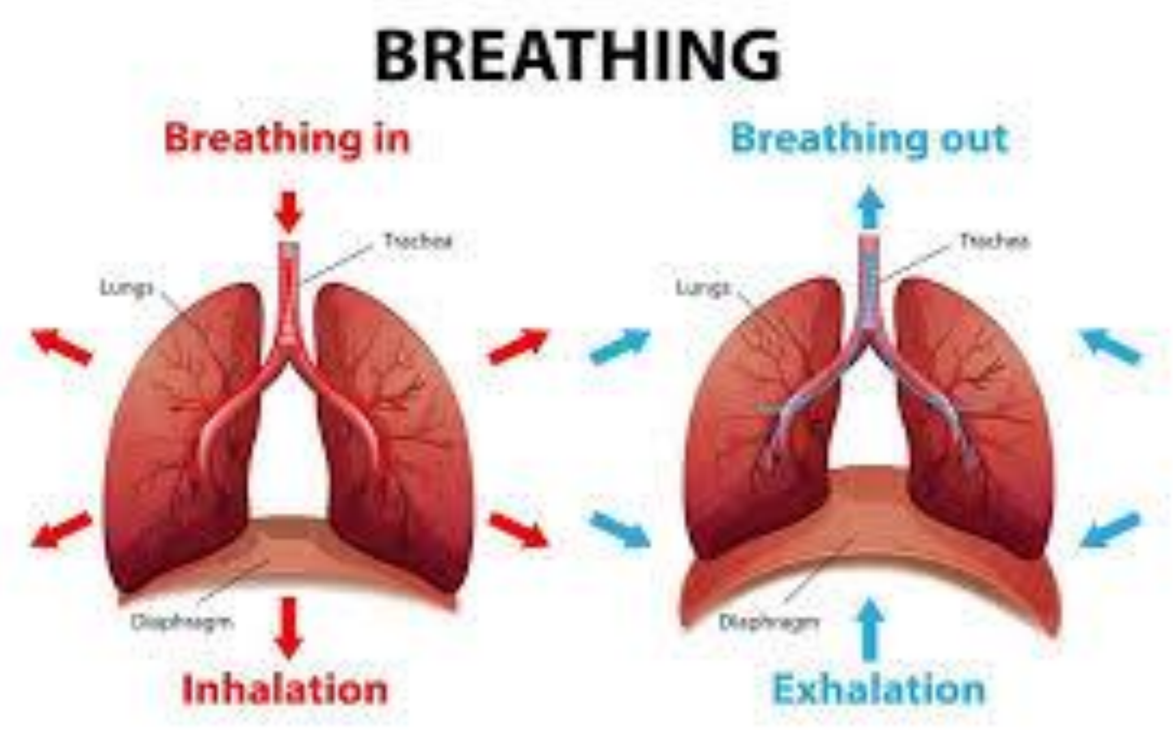
Common Causes

- Occurs over 60% of patients admitted with chest trauma.

Manifestations

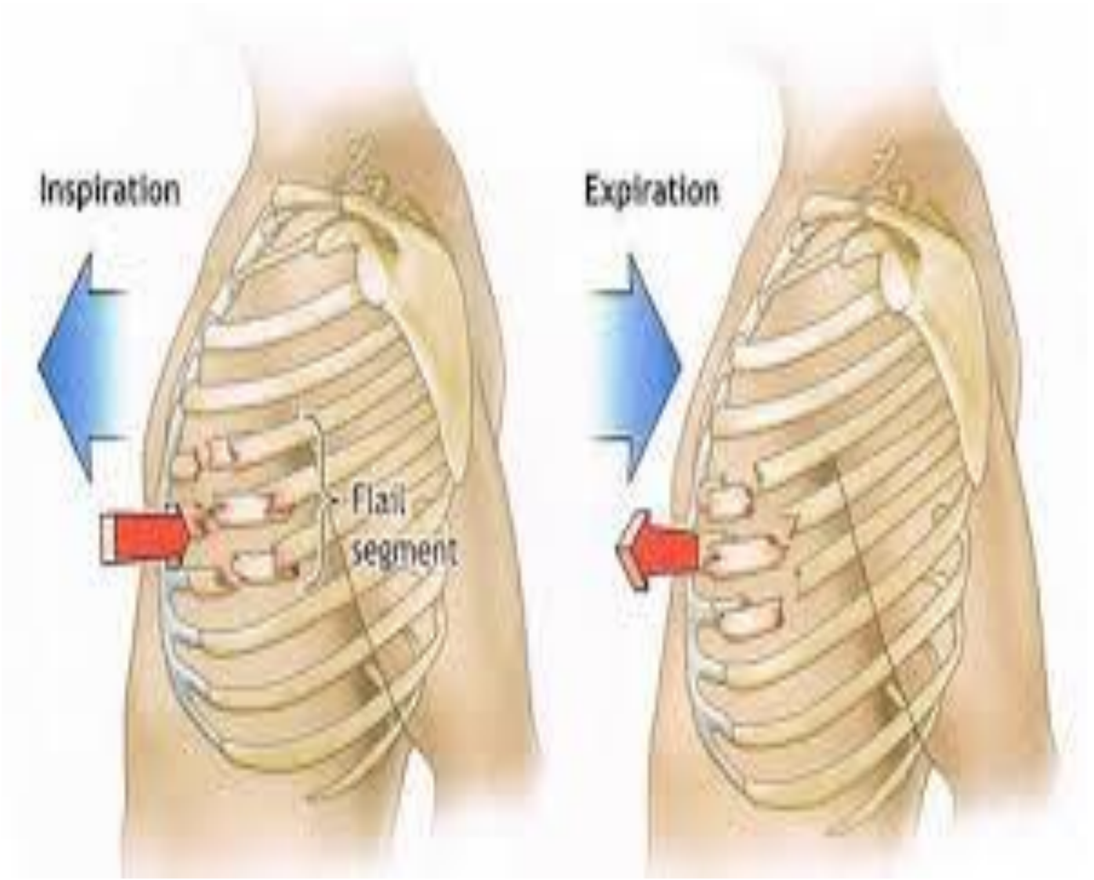
- Fractures of the first three ribs are rare but deadly due to laceration of the subclavian artery.
- Fractures of the lower ribs are associated with injury to the spleen and liver.
- Severe pain
- Bruising
- Tenderness
- Muscle spasm over the site of the fracture.
- A crackling grating sound in the thorax may be detected

Normal breathing cycle



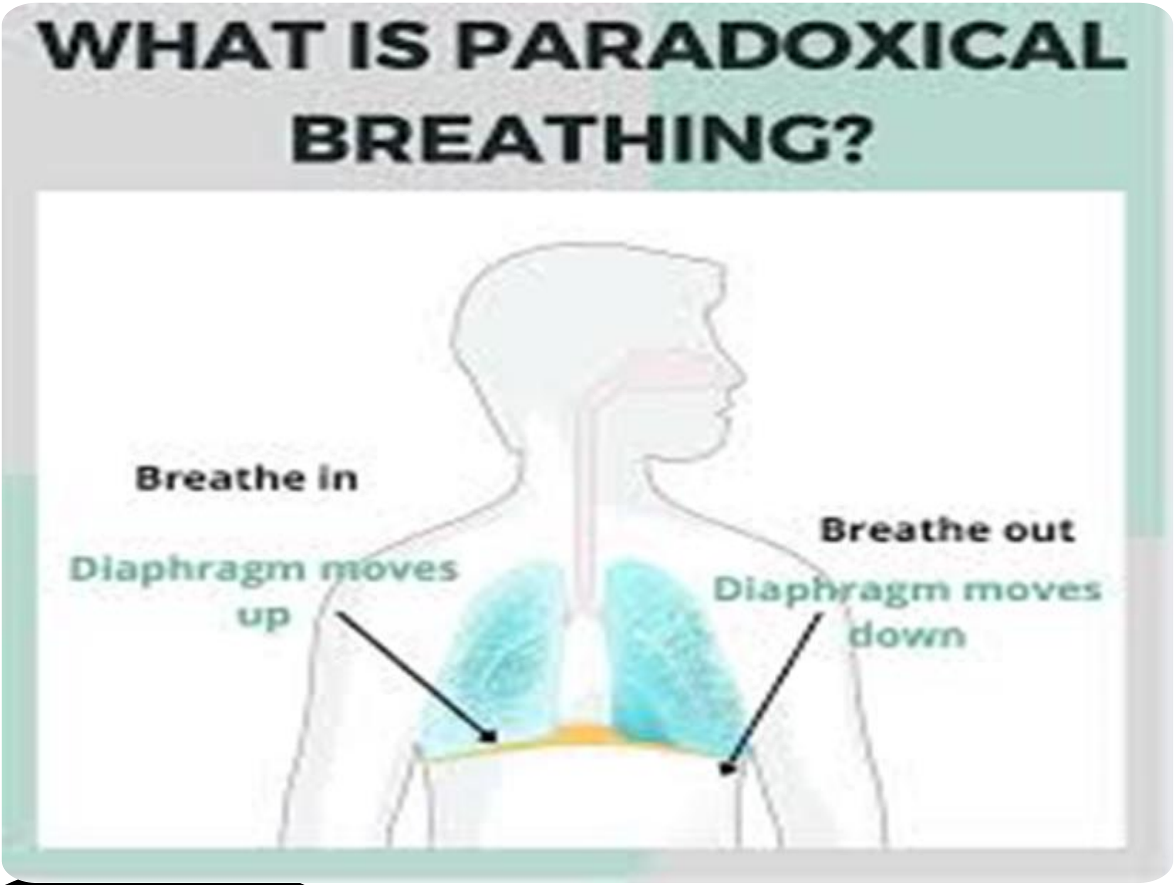
- Normally the chest expands on inspiration, creating a negative intra thoracic pressure; on expiration the chest wall retracts in response to positive pressure

Flail chest



- A severe complication of blunt chest trauma that occurs when **three or more adjacent ribs fracture** in two or more places (or the sternum is fractured, along with ribs adjacent to the sternum fracture). The fracture segment is free of the bony thorax and moves independently in response to intra thoracic pressure.

Paradoxic chest wall



- Paradoxic chest wall motion is the hallmark symptom in diagnosing flail chest. Because a flail chest segment is no longer attached to the bony thorax, it follows the pressure and retracts on inspiration (negative pressure is a pulling pressure) and bulges on expiration (positive pressure is a pushing pressure).

Manifestations of Flail chest

- **During inspiration,** as the chest expands, the detached part of the rib segment (flail segment) will move in a paradoxical manner in that it is pulled inward during inspiration reducing the amount of air that can be taken into the lungs.
- **On expiration**, the flail segment will bulge outward impairing the patient's ability to exhale. Lung contusion and atelectasis often accompany flail chest.

Assessment of chest trauma

Inspection: RR >20 breaths min; hyperpnea; ventilatory distress; use of accessory muscles of respiration; nasal flaring; decreased tidal volume; hemoptysis; asymmetric chest wall motion; paradoxic chest wall motion; inability to clear tracheobronchial secretions; splinting; jugular venous distention; cyanosis or pallor of the skin, lips, and nail beds; and ecchymosis, which can signal injury to underlying organs.

Cont, Assessment of chest trauma

Palpation:

- Tracheal deviation; - subcutaneous emphysema of the neck
- pper portion of the chest; tenderness at fracture points; - flail chest segment;
- weak pulse; - cool, clammy skin;
- protrusion of bony fragments.

Percussion:

- Dullness over lung fields, which can signal hemothorax or atelectasis;
- Hyperresonance over lung fields, signaling pneumothorax.

Cont, Assessment of chest trauma

Auscultation: Diminished or absent breath sounds, respiratory stridor, bony crepitus over fracture sites, muffled heart tones, decreased BP, pericardial friction rub, paradoxic pulse, apical tachycardia. In addition, bowel sounds may be heard in the thorax as a result of rupture or tear of the diaphragm, allowing herniation of abdominal contents into the thorax.

Cont, Chest trauma

2. Penetrating injury: Dyspnea, SOB, moderate chest pain, restlessness, anxiety.

3. Potential complications: Hemothorax, pneumothorax, tension pneumothorax, hemorrhage, shock, and infection.

Diagnostic tests

1. Chest x-ray: Confirms presence of air or fluid in the plcural space; assists in

- determining extent of hemothorax or pneumothorax; and confirms presence or absence of fractures of the bony thorax, as well as mediastinal shift.

2. ABG analysis: Evaluates adequacy of oxygenation and presence or

Cont, diagnosis

3. Pulse oximetry: May be used for continuous monitoring of oxygen saturation.

4. ECG: Reveals presence or absence of life-threatening dysrhythmias and provides a more thorough analysis of the heart's electrical activity. Although dysrhythmias are a common complication after chest trauma, this important test often is overlooked.

5. White blood count: Baseline indicator of an infectious process.

Management of lung injury

1. **Collaborative management:** Interventions are directed toward managing acute respiratory compromise while correcting the underlying path physiology or injuries that complicate or aggravate the patient's condition.

2. **Oxygen therapy:** Delivered by mask or cannula, depending on extent of hypoxemia.

3. **Intubation:** Maintains patent airway, decreases airway resistance and respiratory effort, provides route for easy removal of airway secretions, and allows for manual or mechanical ventilation, as necessary.

Cont, managment

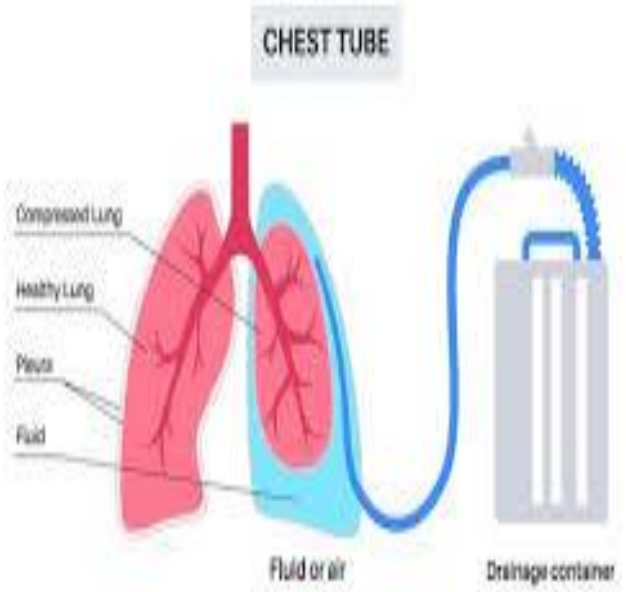
4. Mechanical ventilation: For cases of extreme respiratory distress or ventilator collapse

5. Blood replacement: Generally, blood loss is replaced with packed RBCs or whole fresh blood, when available. Volume usually is replaced with crystalloid fluids (e.g., normal saline, lactatcd Ringer's solution) rather than colloidal IV fluids (e.g., plasma, albumin) because colloids increase the risk of developing acute respiratory distress syndrome (ARDS) and renal failure. In addition, colloids are more expensive and research has failed to demonstrate a significant benefit.

Cont, managment

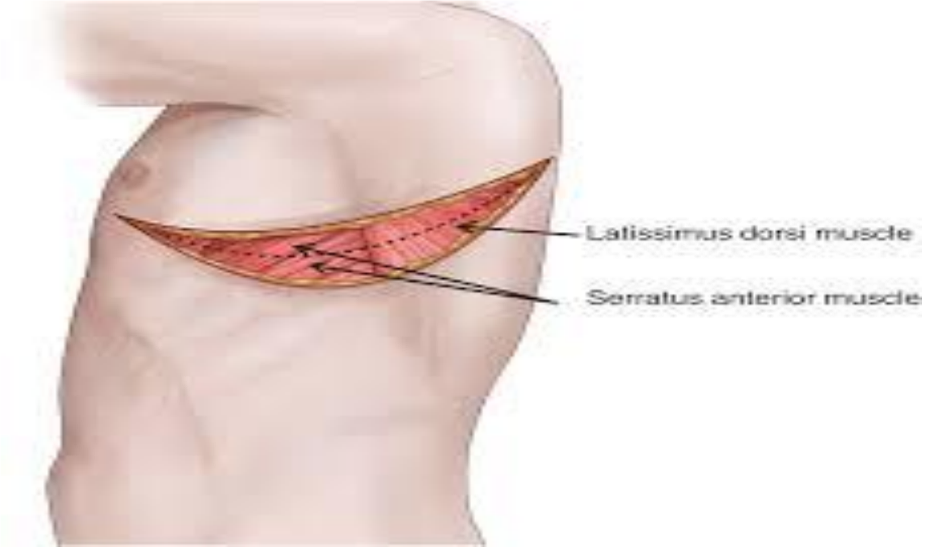
5. Chest tube insertion: To remove accumulation of fluid or air from the chest cavity. Placement depends on the location and extent of the **hemothorax** or pneumothorax and on principles of gravity drainage.

6. Analgesia: Manages pain, which can interfere with work of breathing. Generally, opioid analgesics are avoided or used cautiously because of respiratory depressive side effects. Intercostal nerve block may be performed to provide local pain relief.



7. Stabilization and fixation of flail chest: Most flail chest injuries stabilize within 10-14 days without surgical intervention.

8. Thoracotomy: Generally avoided unless complications develop after stabilization. Indications for thoracotomy include massive air leak in a functioning drainage system, continued or increased bleeding through a chest tube, refractory hypotension, acute deterioration, and cardiac tamponade.



Nursing diagnoses and interventions

Risk for fluid volume deficit related to active loss secondary to excessive bleeding occurring with chest trauma

- **Desired outcome:** Patient is normovolemic as evidenced by BP and HR within patient's normal limits; stable weight; urine output >O.S ml/kg/h; chest drainage <100ml/h; and RR <20 breaths/min with normal depth and pattern .

Management of fluid volume deficit

- **Note condition of dressings and closed chest-drainage system at frequent intervals**. Report excessive drainage or bleeding. Amounts >100 ml/h usually are considered excessive.
- **Record accurate I&O status.**
- **Assess VS at frequent intervals.**

Cont, Management of fluid volume deficit

- **Fluid/Electrolyte Management :** Assess patient's hydration status by monitoring daily weight, as well as amounts of fluid intake and urinary output. Monitor patient's Hgb as an indicator of hemostasis.
- **Airway management and oxygen therapy:** For respiratory monitoring
- **Analgesic Administration:** For pain management
- **Bleeding Reduction:** Blood Products Administration;

Lung injury

Key Terms

- Pneumothorax
- Hemothorax
- Cardiac Tamponade

1) Pneumothorax is an abnormal collection of air in the pleural space between the lung and the chest wall

2) Hemothorax is an accumulation of blood within the pleural cavity

3) Cardiac Tamponade is accumulation of fluid or blood in the pericardium sac around the heart that resulting in compression of the heart.

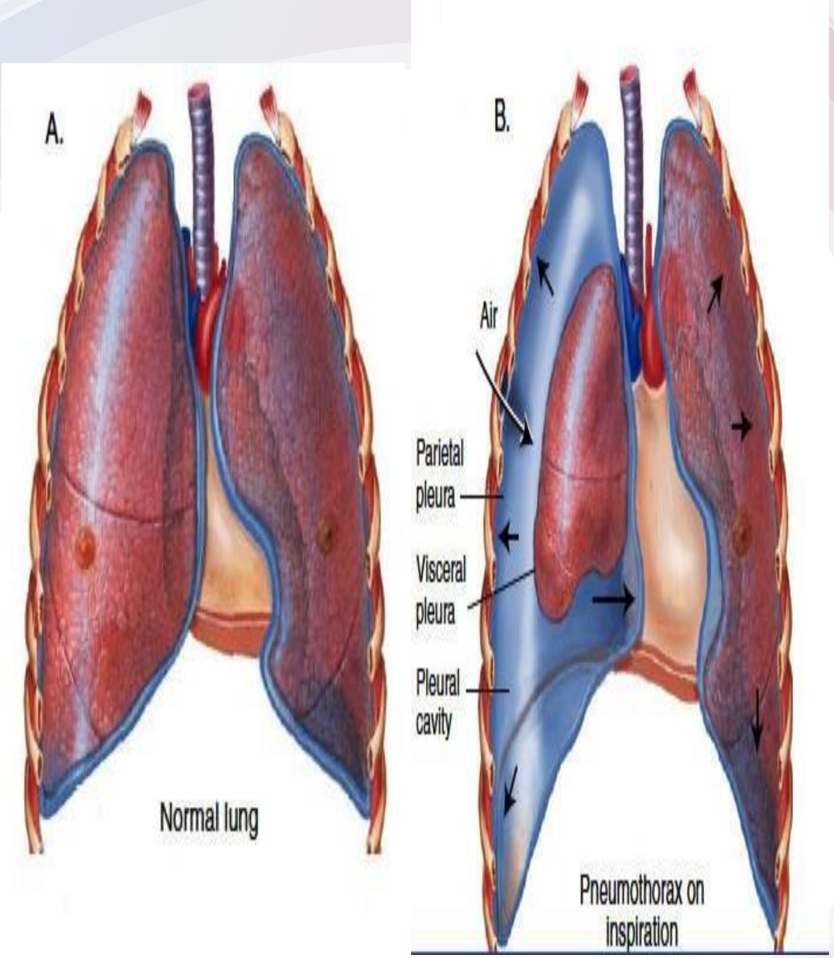
Mechanism of Injury

. **Blunt chest trauma** results from sudden compression or positive pressure directed to the chest wall often by Motor Vehicle Accident (MVA) or falls.

- **Penetrating trauma** occurs when a foreign object penetrates the chest wall such as low-velocity weapons (gun, knife). Stab wounds that involve the anterior chest wall.

Pneumothorax

Pneumothorax occurs when the parietal or visceral pleura are breached and the pleural space is exposed to positive atmospheric pressure. Normally the pressure in the pleural space is negative or sub atmospheric; this negative pressure is required to maintain lung inflation. When pleura are breached, air enters the pleural space, and the lung or a portion of it collapses.



Types of Pneumothorax

1. **A simple or spontaneous pneumothorax:** Occurs as air enters the pleural space through the rupture of a bleb or a bronchopleural fistula.

Causes: It may occur in an apparently healthy person in the absence of trauma due to rupture of an air filled bleb, or blister, on the surface of the lung, allowing air from the airways to enter the pleural cavity. It may be associated with diffuse interstitial lung disease and severe emphysema.

Cont, Types of Pneumothorax

2. A traumatic pneumothorax: Occurs when air escapes from a laceration in the lung itself and enters the pleural space or from a wound in the chest wall.

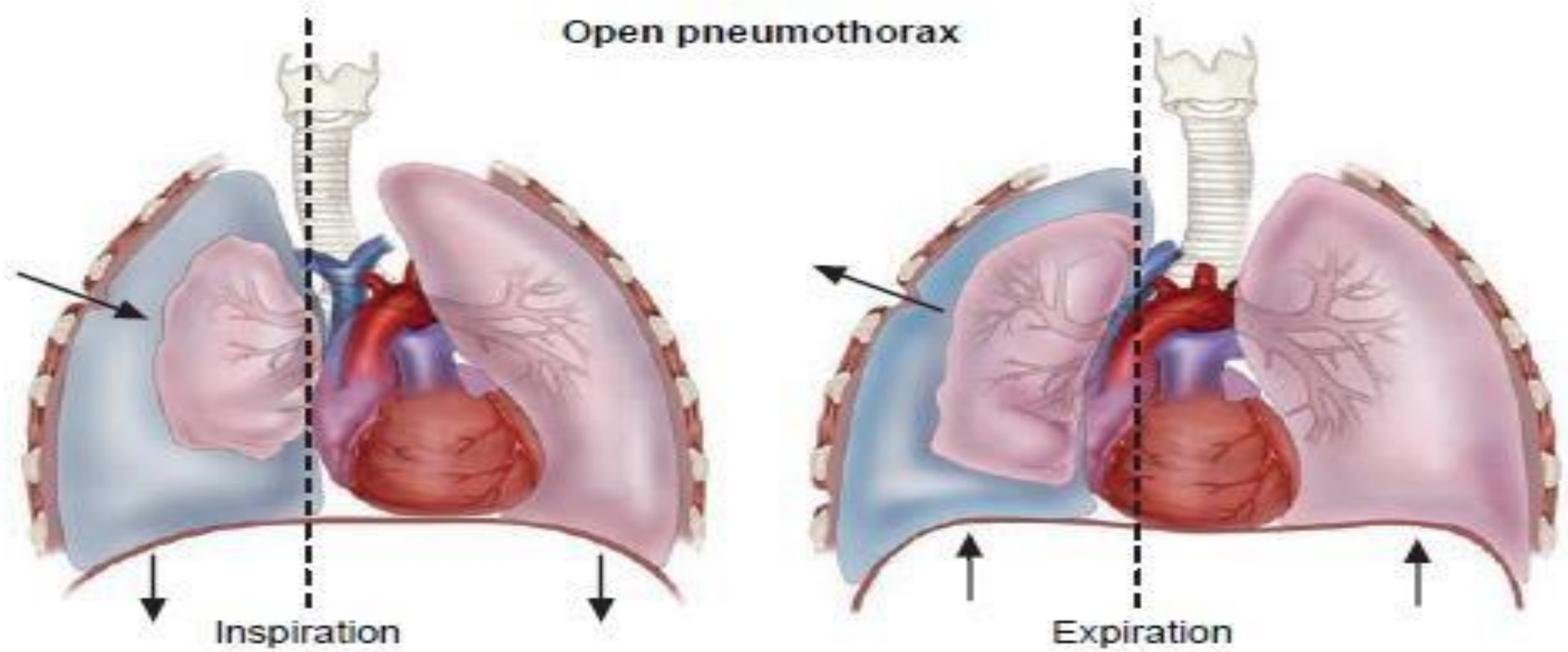
Causes:

- Blunt trauma as rib fractures, penetrating chest
- Abdominal trauma as stab wounds or gunshot wounds, or diaphragmatic tears. Traumatic pneumothorax may occur during invasive thoracic procedures as thoracentesis and biopsy.

Cont, Types of Pneumothorax

3. Open pneumothorax is one form of traumatic pneumothorax. It occurs when a wound in the chest wall is large enough to allow air to pass freely in and out of the thoracic cavity with respiration that produces a sucking sound. Results not only the lung collapse, but the heart and great vessels also shift toward the uninjured side with each inspiration and in the opposite direction with expiration. This is termed mediastinal flutter, it produces serious circulatory problems.

Open pneumothorax



Cont,Types of Pneumothorax

4. A tension pneumothorax: Occurs when air is drawn into the pleural space from a lacerated lung or through a small opening or wound in the chest wall. Air flows into the pleural space with inspiration and becomes trapped, it cannot be expelled during expiration through the air passages or the opening in the chest wall. In effect, *a one-way valve mechanism* occurs where air enters the pleural space but cannot escape.

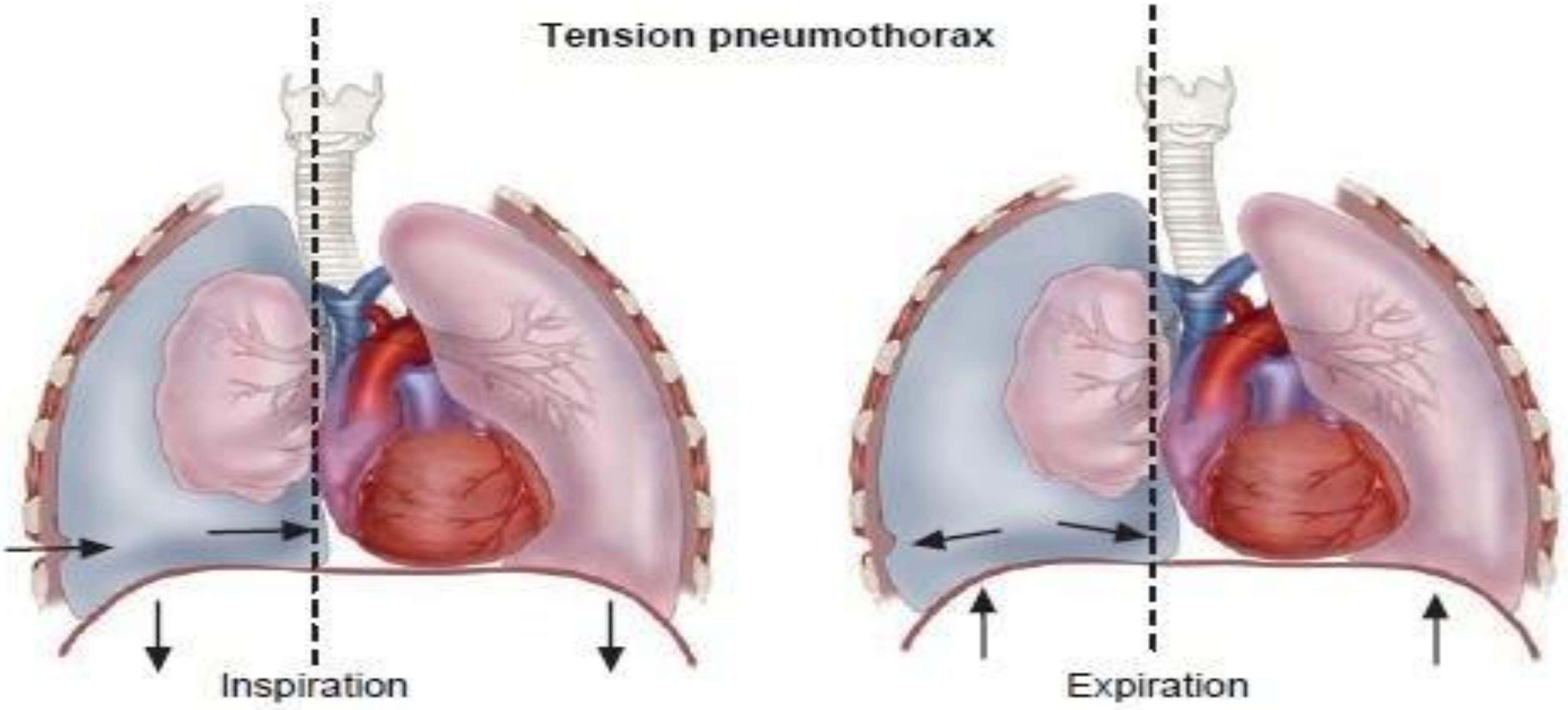
Tension pneumothorax

With each breath, tension (positive pressure) is increased within the affected pleural space. This causes the lung to collapse and the heart, the great vessels, and the trachea to shift toward the unaffected side of the chest *(mediastinal shift to the opposite side).*

As pressure continues to build, the shift exerts pressure on the heart and thoracic aorta, which results in decreased venous return and decreased cardiac output. Tissue perfusion with oxygenated blood is further impaired because the collapsed lung cannot participate in gas exchange.

Impairment of peripheral circulation and the pulse may be undetectable this is known as ***pulseless electrical activity.***

Tension pneumothorax



Clinical Manifestations

- Pleuritic pain is usually sudden.
- The patient may have only minimal respiratory distress with slight chest discomfort
- Tachypnea
- Small simple or uncomplicated pneumothorax.
- **If the pneumothorax is large and the lung collapses** totally, acute respiratory distress occurs. The patient is anxious, has dyspnea and air hunger, has increased use of the accessory muscles, and may develop central cyanosis from severe hypoxemia

Cont, Manifestations

- **In *simple pneumothorax***
- Trachea is midline
- Expansion of the chest is decreased
- Breath sounds may be diminished
- Percussion of the chest may reveal normal sounds or hyperresonance depending on the size of the pneumothorax.

Cont, Manifestations

- **In *tension pneumothorax***
- Air hunger, agitation, increasing hypoxemia, central cyanosis, hypotension, tachycardia, and profuse diaphoresis.
- The trachea is shifted away from the affected side
- Chest expansion may be decreased or fixed in a hyperexpansion state
- Breath sounds are diminished or absent, and percussion to the affected side is hyperresonant.

Medical Management

1. The goal of treatment is to evacuate the air or blood from the pleural space.

2. A small chest tube is inserted near the second intercostal space.

3. If a patient also has a hemothorax, a large diameter chest tube is inserted, usually in the fourth or fifth intercostal space at the midaxillary line.

4. Once the chest tube is inserted and suction is applied (usually to 20 mm Hg suction), effective decompression of the pleural cavity (drainage of blood or air) occurs.

5.The pleural cavity can be decompressed by needle aspiration (*thoracentesis*) or by chest tube drainage of the blood or air.

6. The chest wall is opened surgically (*thoracotomy*) if more than 1500 mL of blood is aspirated initially by thoracentesis or if the initial chest tube output or if chest tube output continues at greater than 200 mL/h.

- **The patient with a possible *tension pneumothorax***

1. Should immediately be given a high concentration of supplemental oxygen to treat the hypoxemia
2. Pulse oximetry should be used to monitor oxygen saturation.

In an emergency situation

1. Inserting a large bore needle (14- gauge) at the second intercostal space, midclavicular line on the affected side for converting a tension pneumothorax to a simple pneumothorax.
2. If the lung re-expands and air leakage from the lung parenchyma stops
3. If a prolonged air leak continues from chest tube drainage to underwater seal, surgery may be necessary to close the leak.

Guidelines for Chest Tube Drainage Systems

1. If using a chest drainage system with a water seal, fill the water seal chamber with sterile water to the level specified by the manufacturer.

2. When using suction in chest drainage systems with a water seal, fill the suction control chamber with sterile water to the 20-cm level or as prescribed. In systems without a water seal, set

3. Attach the drainage catheter exiting the thoracic cavity to the tubing coming from the collection chamber. Tape securely with adhesive tape.

4. If suction is used, connect the suction control chamber tubing to the suction unit.

5. Mark the drainage from the collection chamber with tape on the outside of the drainage unit

6. Ensure that the drainage tubing does not kink, loop, or interfere with the patient's movements.

7. Encourage the patient to assume a comfortable position with good body alignment. With the lateral position, make sure that the patient's body does not compress the tubing. The patient should be turned and repositioned every 1.5 to 2 hours. Provide adequate analgesia.

8. Assist the patient with range-of-motion exercises for the affected arm and shoulder several times daily. Provide adequate analgesia.

9. Gently “milk” the tubing in the direction of the drainage chamber as needed.

10. Make sure there is fluctuation (tidaling) of the fluid level in the water seal chamber (in wet systems), or check the air leak indicator for leaks (in dry systems with a one-way valve).

11. When assisting in the chest tubes removal, instruct the patient to perform a gentle Valsalva maneuver or to breathe quietly.

- Note: Fluid fluctuations in the water seal chamber or air leak indicator area will stop when:
- The lung has re-expanded
  - The tubing is obstructed by blood clots, fibrin, or kinks

- A loop of tubing hangs below the rest of the tubing
- Suction motor or wall suction is not working properly
- Observe for air leaks in the drainage system; they are indicated by constant bubbling in the water seal chamber, or by the air leak indicator in dry systems with a one-way valve
- Encourage the patient to breathe deeply and cough at frequent intervals

Hemothorax

Hemothorax results from blunt or penetrating thoracic trauma that can cause bleeding into the pleural space, resulting in a hemothorax.

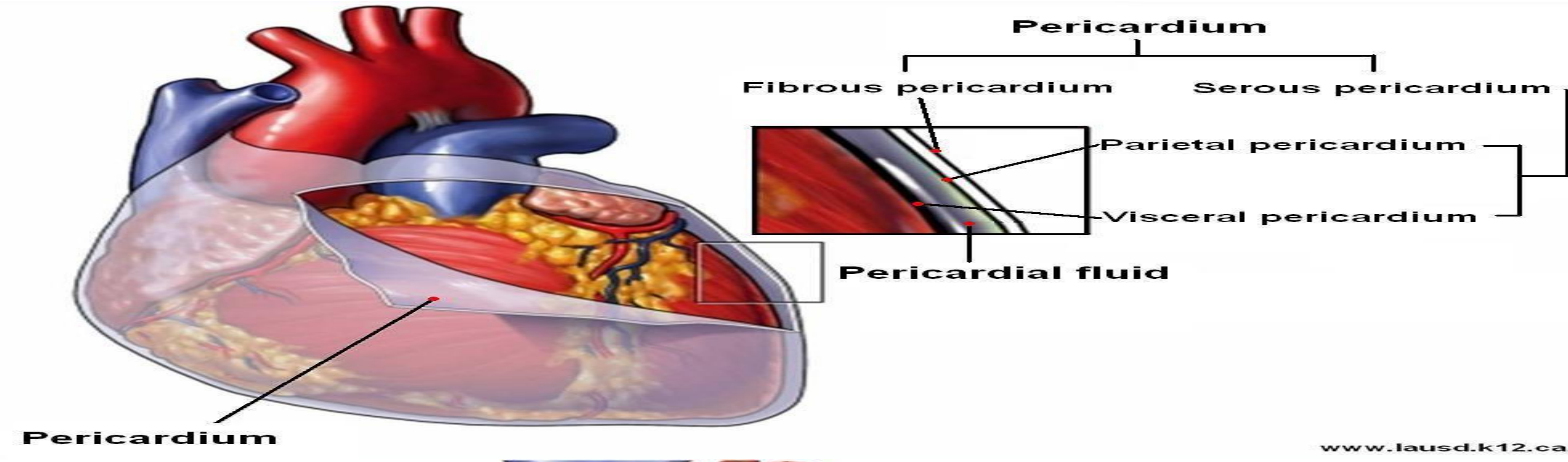
A massive hemothorax results from the accumulation of more than 1500 mL of blood in the chest cavity.The source of bleeding may be the intercostal or internal mammary arteries, lungs, heart, or great vessels.

Clinical manifestation of hemothorax

- Manifestation of patients with a hemothorax includes manifestation of hypovolemic shock.
- Breath sounds may be diminished or absent over the affected lung.
- With hemothorax, the neck veins are collapsed, and the trachea is at midline.

Cont, Manifestation of hemothorax

Massive hemothorax can be diagnosed on the basis of clinical manifestations of hypotension associated with the absence of breath sounds or dullness to percussion on one side of the chest.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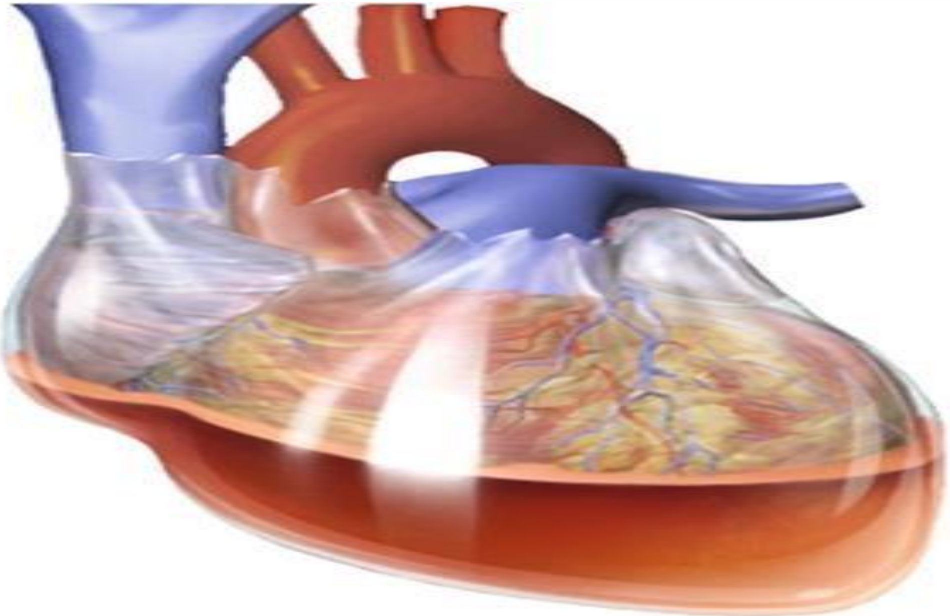

Medical Management

- Resuscitation with intravenous fluids is initiated to treat the hypovolemic shock.
- A chest tube is placed on the affected side to allow drainage of blood.
- Thoracotomy may be necessary for patients who require persistent blood transfusions or who have significant bleeding (200 mL/ hour for 2 to 4 hours or more than 1500 mL on the initial tube insertion) or when there are injuries to major cardiovascular structures.

Cardiac Tamponade

Definition: Is the progressive accumulation of blood in the pericardial sac. With cardiac tamponade, progressive accumulation of 120 to 150 mL of blood increases the intracardiac pressure and compresses the atria and ventricles.

- An increase in intracardiac pressure leads to decreased venous return and decreased filling pressure, which leads to decreased cardiac output, myocardial hypoxia, cardiac failure, and cardiogenic shock.

Manifestation of cardiac tamponade

1. Presence of elevated central venous pressure with neckvein distention (Beck's triad)

2. Muffled heart sounds

3. Hypotension.

4. Pulseless electrical activity (PEA) in the absence of hypovolemia and tension pneumothorax suggests cardiac tamponade.

5. Bedside ultrasound is used to diagnose cardiac tamponade

Medical Management of cardiac tamponade

1. Immediate treatment is required to remove the accumulation of fluid in the pericardial sac.

2. *Pericardiocentesis* involves aspiration of fluid from the pericardium by use of a large-bore needle. The inherent risk in this procedure is potential laceration of the coronary artery.

3. Other approaches include surgical procedures such as *thoracotomy.* The goal of these procedures is to locate and control the source of bleeding.